

CAPSTONE PROJECT REPORT

Project Name: Nykaa

Module Name: Skin

Superset Number: 4908170

Domain: E-Commerce

Submitted By: Harsh Singh

Project Synopsis

The Nykaa Automation Testing Project is an end-to-end automation framework developed for testing the functionality and usability of the Nykaa E-Commerce platform. The project focuses on validating important user workflows such as login functionality, product search, cart management, checkout process, and payment flow.

The framework was designed using Selenium with Python and later enhanced using BDD (Behavior Driven Development) concepts for better readability, maintainability, and collaboration.

The main objective of this project is to reduce manual testing effort, improve testing efficiency, identify defects quickly, and ensure that critical functionalities of the application work correctly across different scenarios.

Project Objectives

- Automate functional testing of the Nykaa website
- Perform end-to-end testing of major workflows
- Validate positive and negative test scenarios
- Improve test execution speed and reliability
- Generate detailed execution reports using Allure
- Implement reusable and scalable automation framework using POM and BDD

Modules Tested

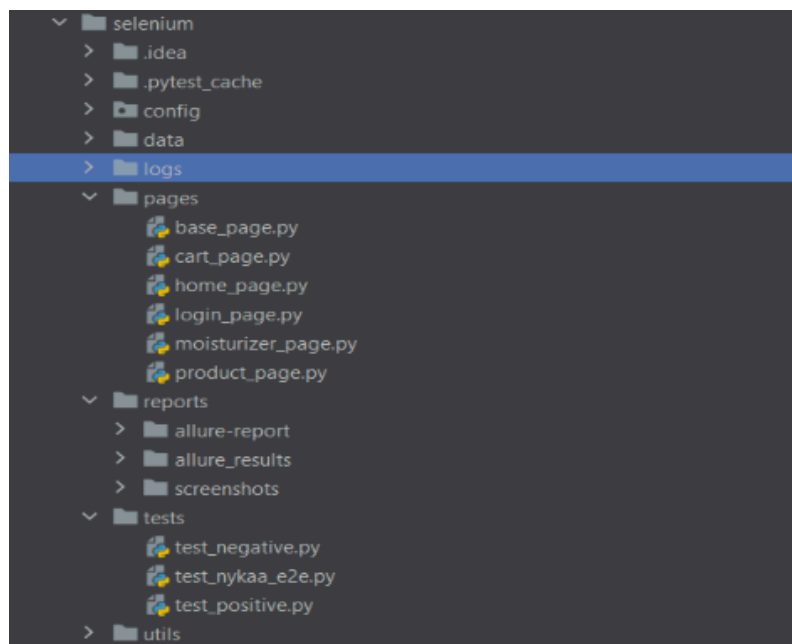
- Login Module
- Product Search Module
- Product Selection
- Cart Functionality
- Checkout Process
- Positive Test Scenarios
- Negative Test Scenarios
- End-to-End User Journey

Technologies / Tech Stack Used

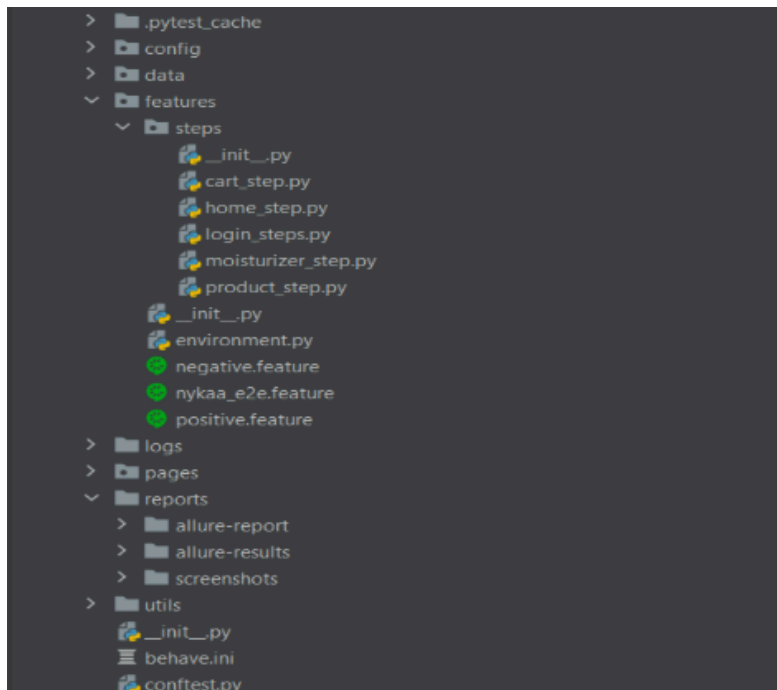
- **Python** - Primary programming language
- **Selenium WebDriver** - Web automation tool
- **Pytest** - Testing framework
- **behave** - BDD framework
- **Allure Reporting** - Test reporting tool
- **Execution Logs** - Detailed logging system
- **GitHub** - Version control system
- **Page Object Model (POM)** - Design pattern
- **Behavior Driven Development (BDD)** - Testing methodology

Test Strategy

The project followed an automation-first testing approach to ensure efficient, reliable, and scalable validation of critical business workflows. Automated test scripts were developed using Python and Selenium WebDriver following the Page Object Model (POM) design pattern for better maintainability, reusability, and modularity.



Behavior-Driven Development (BDD) methodology was implemented to enhance the readability of test scenarios and improve collaboration between technical and non-technical stakeholders. Test cases were written in a business-readable format using Gherkin syntax (Given-When-Then).



The automation framework was integrated with various supporting tools and technologies to enable streamlined test execution, reporting, and continuous integration.

Testing Scope

The following testing types were performed during the project:

- **End-to-End Testing**

Validated complete user workflows from login to checkout to ensure the application functions correctly as a whole.

- **Positive Testing**

Verified application behavior using valid inputs and expected user actions.

- **Negative Testing**

Tested application stability and validation handling using invalid inputs and unexpected scenarios.

Framework Architecture

The framework was developed using Page Object Model (POM) and Behavior Driven Development (BDD). POM helped separate locators from test logic for better maintainability and scalability, while BDD improved readability of scenarios.

Test Environment

Component	Details
Operating System	Windows
Browser	Google Chrome
IDE	Pycharm
Programming Language	Python
Automation Tool	Selenium WebDriver
Framework	Pytest & behave
Reporting Tool	Allure Report & Execution Logs
Version Control	GitHub

Test Execution

The test cases were executed using Pytest and BDD commands from the terminal.

Pytest Command:

```
pytest -v
```

BDD Command:

```
behave -v
```

Allure Report Command:

```
allure serve allure-results
```

Test Output

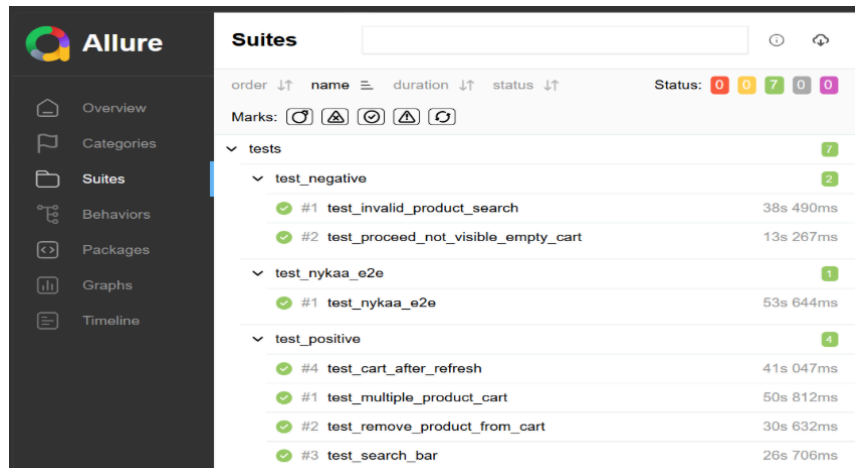
Total Test Cases: 7

- End-to-End Test Cases: 1
- Positive Test Cases: 4
- Negative Test Cases: 2

Selenium Allure Report

The Selenium automation framework generated comprehensive Allure reports showing detailed test execution results, including:

- Test suite overview with pass/fail status
- Individual test case execution times
- Detailed step-by-step execution logs
- Screenshots for test cases



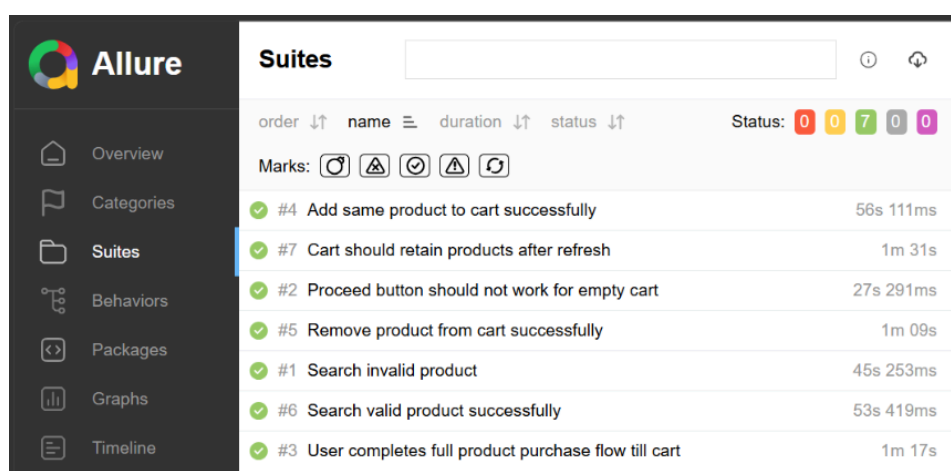
The screenshot shows the Allure web interface. On the left is a sidebar with navigation links: Overview, Categories, Suites (selected), Behaviors, Packages, Graphs, and Timeline. The main area is titled 'Suites' and displays a table of test results. The table has columns for 'order', 'name', 'duration', and 'status'. A status bar at the top right shows counts: 0 red, 0 yellow, 7 green, 0 blue, 0 purple. The table lists several test suites, each with a green checkmark icon and a count in a green box. The 'tests' suite has 7 tests, 'test_negative' has 2, 'test_nykaa_e2e' has 1, and 'test_positive' has 4. Each test entry includes a green checkmark, a number, the test name, and its duration.

order	name	duration	status
Status: 0 0 7 0 0			
Marks: [Icons]			
✓	tests		7
✓	test_negative		2
✓ #1	test_invalid_product_search	38s 490ms	
✓ #2	test_proceed_not_visible_empty_cart	13s 267ms	
✓	test_nykaa_e2e		1
✓ #1	test_nykaa_e2e	53s 644ms	
✓	test_positive		4
✓ #4	test_cart_after_refresh	41s 047ms	
✓ #1	test_multiple_product_cart	50s 812ms	
✓ #2	test_remove_product_from_cart	30s 632ms	
✓ #3	test_search_bar	26s 706ms	

BDD Allure Report

The BDD framework provided business-readable test reports with:

- Feature-based test organization
- Scenario execution details
- Step-by-step validation results
- Clear pass/fail indicators for each test scenario



The screenshot shows the Allure web interface for a BDD report. The sidebar is the same as the previous screenshot. The main area is titled 'Suites' and displays a table of test results. The status bar shows counts: 0 red, 0 yellow, 7 green, 0 blue, 0 purple. The table lists several test scenarios, each with a green checkmark icon and a count in a green box. The scenarios are: #4 Add same product to cart successfully (56s 111ms), #7 Cart should retain products after refresh (1m 31s), #2 Proceed button should not work for empty cart (27s 291ms), #5 Remove product from cart successfully (1m 09s), #1 Search invalid product (45s 253ms), #6 Search valid product successfully (53s 419ms), and #3 User completes full product purchase flow till cart (1m 17s).

order	name	duration	status
Status: 0 0 7 0 0			
Marks: [Icons]			
✓ #4	Add same product to cart successfully	56s 111ms	
✓ #7	Cart should retain products after refresh	1m 31s	
✓ #2	Proceed button should not work for empty cart	27s 291ms	
✓ #5	Remove product from cart successfully	1m 09s	
✓ #1	Search invalid product	45s 253ms	
✓ #6	Search valid product successfully	53s 419ms	
✓ #3	User completes full product purchase flow till cart	1m 17s	

Execution Logs

Detailed execution logs were generated for both Selenium and BDD frameworks, providing:

- **Selenium Logs:** Timestamped actions including page loads, element interactions, assertions, and navigation steps

```
2026-05-22 12:40:35,889 | INFO | START TEST: test_nykaa_e2e
2026-05-22 12:40:35,946 | INFO | LOAD | Test data loaded from Excel successfully
2026-05-22 12:40:35,948 | INFO | DATA | BRAND_NAME = Cetaphil
2026-05-22 12:40:42,295 | INFO | OPEN | Nykaa website launched
2026-05-22 12:41:21,432 | INFO | ASSERT | Homepage loaded successfully
2026-05-22 12:41:21,440 | INFO | CLICK | Sign In button clicked
2026-05-22 12:41:26,956 | INFO | SELECT | Mobile / Email option selected
2026-05-22 12:41:43,810 | INFO | CLICK | Skip login clicked
2026-05-22 12:42:03,301 | INFO | HOVER | Hovering on Skin menu
2026-05-22 12:42:04,012 | INFO | CLICK | Clicking Moisturizer category
2026-05-22 12:42:17,027 | INFO | ASSERT | Moisturizer page validated
2026-05-22 12:42:17,030 | INFO | SELECT | Brand selected from Excel: Cetaphil
2026-05-22 12:42:22,001 | INFO | CLICK | First product selected
2026-05-22 12:42:34,194 | INFO | ASSERT | Product page validated
2026-05-22 12:42:34,196 | INFO | ACTION | Product added to bag
2026-05-22 12:42:38,781 | INFO | NAVIGATE | Cart opened
2026-05-22 12:42:46,015 | INFO | ACTION | Proceed button clicked
2026-05-22 12:43:16,519 | INFO | E2E TEST COMPLETED SUCCESSFULLY
```

- **BDD Logs:** Scenario-based logging with detailed step execution and validation results

```
2026-05-23 12:20:30,929 | INFO | START SCENARIO: User completes full product purchase flow till cart
2026-05-23 12:20:30,935 | INFO | Opening Nykaa website
2026-05-23 12:20:30,936 | INFO | Initializing Home Page
2026-05-23 12:20:44,367 | INFO | Nykaa website opened successfully
2026-05-23 12:20:53,166 | INFO | Opening login popup
2026-05-23 12:20:53,197 | INFO | Login Page initialized
2026-05-23 12:21:04,666 | INFO | Login popup opened successfully
2026-05-23 12:21:08,293 | INFO | Selecting mobile or email login option
2026-05-23 12:21:11,168 | INFO | Mobile or email login option selected successfully
2026-05-23 12:21:12,065 | INFO | Clicking skip login button
2026-05-23 12:21:26,354 | INFO | Skip login button clicked successfully
2026-05-23 12:21:32,755 | INFO | Hovering on skin menu
2026-05-23 12:21:34,304 | INFO | Hovered on skin menu successfully
2026-05-23 12:21:35,354 | INFO | Clicking moisturizers category
2026-05-23 12:21:37,451 | INFO | Switching to moisturizer tab
2026-05-23 12:21:37,621 | INFO | Moisturizer category opened successfully
2026-05-23 12:21:51,062 | INFO | Loading brand data from Excel
2026-05-23 12:21:51,107 | INFO | Brand loaded from Excel: Cetaphil
2026-05-23 12:21:51,109 | INFO | Moisturizer Page initialized
2026-05-23 12:21:52,533 | INFO | Brand selected successfully: Cetaphil
2026-05-23 12:21:53,194 | INFO | Selecting first moisturizer product
2026-05-23 12:21:53,197 | INFO | Moisturizer Page initialized
2026-05-23 12:21:54,979 | INFO | First moisturizer product selected
2026-05-23 12:21:55,066 | INFO | Switched to product page tab successfully
2026-05-23 12:22:18,282 | INFO | Adding product to bag
2026-05-23 12:22:18,536 | INFO | Product Page initialized
2026-05-23 12:22:20,624 | INFO | Product added to bag successfully
2026-05-23 12:22:30,546 | INFO | Opening cart from product page
2026-05-23 12:22:30,656 | INFO | Product Page initialized
2026-05-23 12:22:33,803 | INFO | Cart opened successfully from product page
```

Test Conclusion

The Nykaa Automation Testing Project successfully automated major functionalities of the Nykaa E-Commerce application using Selenium, Python, Pytest, and BDD.

The framework validated critical workflows such as login, search, cart operations, checkout, and payment processing. Positive and negative test cases were executed successfully to ensure application stability and reliability.

The implementation of POM improved maintainability of the framework, while BDD enhanced readability and collaboration. Allure reporting provided detailed execution insights and failure analysis through screenshots and reports.

Overall, the project demonstrated how automation testing can improve software quality, reduce manual effort, and accelerate testing processes in real-world E-Commerce applications.